MA 214 APPLIED STATISTICS · PROJECT 1 · GROUP C1G1

Signs of Mental Illness in Fatal Police Shootings

Monica Chen Breck Johnson Nico Lehtinen Larianna Ramsey

Research Question

WHY IT MATTERS

- US police fatally shoot roughly 1,000 people a year, and about **1 in 4** showed signs of mental illness. That fuels the debate over sending trained crisis responders instead of armed officers.
- Knowing which circumstances go with mental-health cases helps describe who these encounters involve.

PRIMARY

How are **weapon type** and **fleeing status** associated with whether the person showed signs of mental illness?

SECONDARY

After accounting for those, do **age** and **threat level** add anything?

Hypotheses: fleeing → lower odds; knives and other non-gun objects → higher odds than guns.

Data and Cleaning Decisions

Source: Washington Post *Fatal Force* database (OpenIntro copy)
Collected: reporters compile each case from local news, police websites and social media
Population: 6,421 people shot and killed by on-duty US police, Jan 2015 – Jul 2021; one row per person
Type: observational, close to a census of all cases (not a random sample)

Variable	Role	Description
signs_of_mental_illness	Response	1 if news reports noted mental illness
armed	Predictor	Weapon, 5 groups (gun = reference)
flee	Predictor	Not fleeing (reference), car, foot, other
age	Predictor	Age in years
threat_level	Predictor	Attack (reference), other, undetermined

Data cleaning:

- **Duplicates:** removed 2 exact duplicate rows (no ID column to tell them apart)
- **Weapon groups:** ~90 labels grouped into gun, knife/sharp, unarmed, vehicle, other; toy and BB guns → other
- **Missing values:** unknown weapon (457), unknown fleeing (434) or missing age (285) kept for EDA, dropped from the model
- **No imputation:** missing values are likely not random (an "undetermined" weapon is itself informative)

Final model data: 5,371 people
1,048 rows removed

The Response and Age

SIGNS OF MENTAL ILLNESS

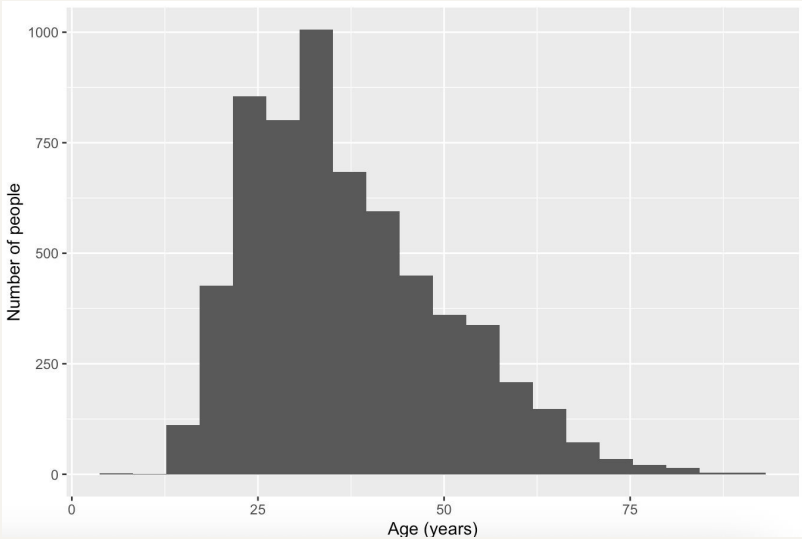
24%

1,315 of 5371 people. Unbalanced but with plenty of “yes” cases for logistic regression.

AGE (YEARS)

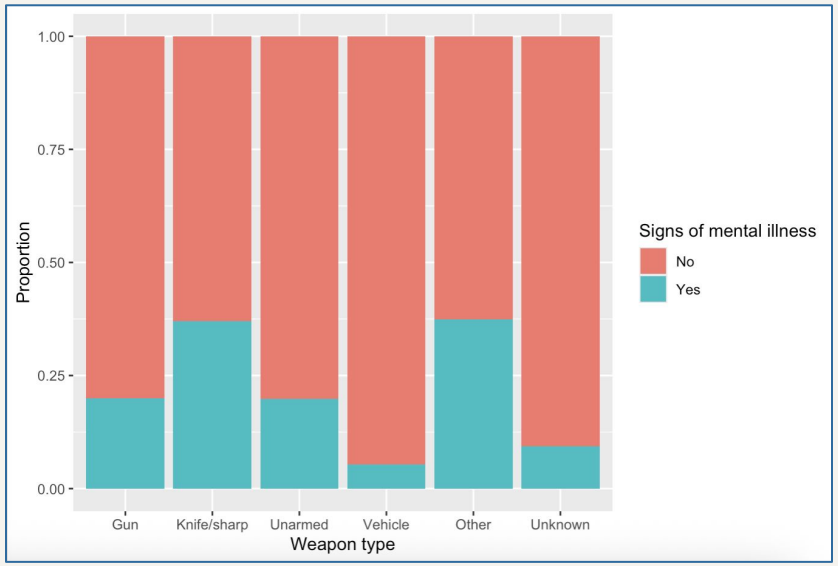
Group	Signs of mental illness	No signs
Median	37	34
Mean	39.7	36.3
Range	15–89	6–91
Middle 50%	29–49	27–45
Skew	Right	Right (slightly more)

AGE DISTRIBUTION

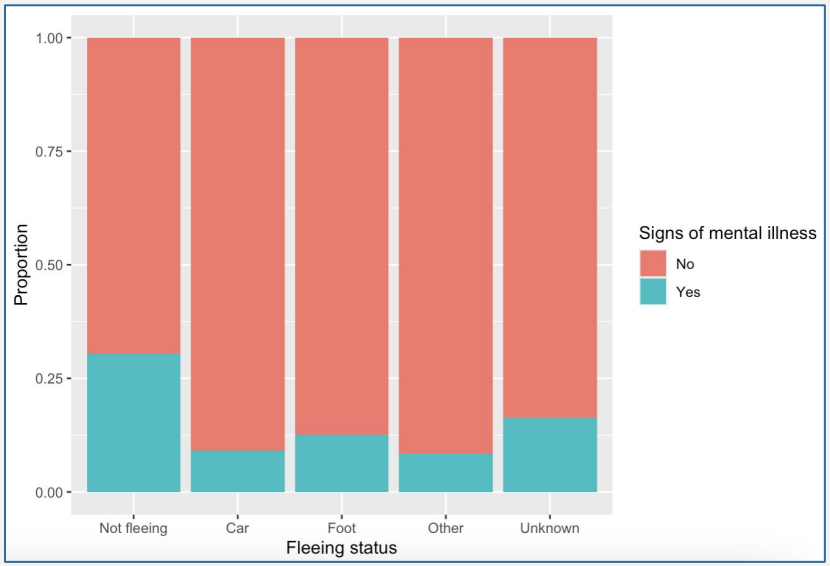


Share Showing Signs of Mental Illness

BY WEAPON



BY FLEEING STATUS



Both patterns point the way we hypothesized. Threat level barely moves the rate: 22% for “attack” vs 25% for “other.”

Candidate Models

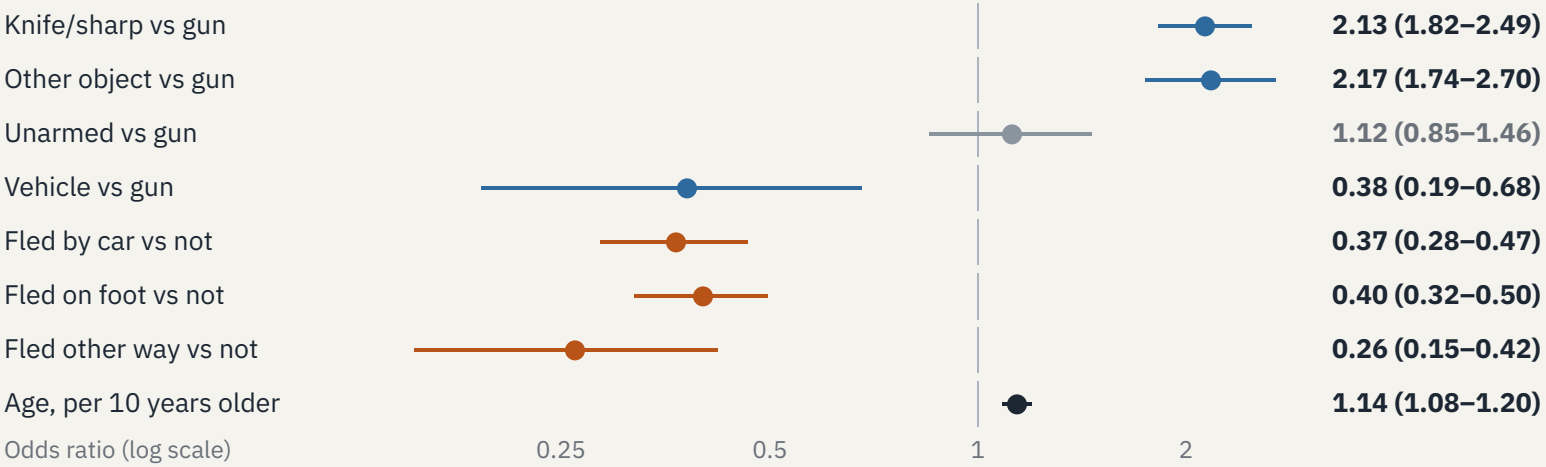
Logistic model (n = 5,371)	AIC	Accuracy
M1: response = age	5922.8	0.755
M2: response = age + weapon	5713.0	0.754
M3: response = age + weapon + flee (chosen)	5567.1	0.755
M4: response = age + weapon + flee + threat	5569.4	0.754

Weapon type and fleeing matter, causing a 210 and 146 decrease in AIC respectively.

Threat level isn't significant, as it increases AIC when added as a predictor

Pick M3: age +weapon+ flee
Lowest AIC; accuracy can't separate the models

Final Model: Odds Ratios



Holding the others fixed, a knife roughly **doubles** the odds of reported mental illness compared with a gun; fleeing in any way cuts the odds by **60–74%**.

From Log Odds to Probability

STEP 1: LOG ODDS

$$\text{log-odds} = -1.592 + 0.0131 \cdot \text{age} + 0.757 \cdot \text{knife} + \ldots - 1.006 \cdot \text{car} - 0.916 \cdot \text{foot}$$

Each category term is 1 if it applies, 0 if not. Gun and not fleeing are the reference levels.

STEP 2: PROBABILITY

$$\hat{p} = e^{(\text{log-odds})} \div (1 + e^{(\text{log-odds})})$$

What predict (type = "response") does in R.

Not fleeing, age …	Log odds	e^(log odds)	Predicted prob.
Gun, age 25	−1.592 + 0.0131×25 = −1.264	0.282	22.0%
Gun, age 45	−1.592 + 0.0131×45 = −1.002	0.367	26.8%
Knife, age 25	−1.264 + 0.757 = −0.507	0.602	37.6%

Same age, same fleeing status: switching from a gun to a knife raises the predicted chance from 22% to 38%.

Predicted Probabilities

Two people who had a gun and were not fleeing. Only age differs.

AGE 25

22.0%

predicted chance of signs of mental illness

AGE 45

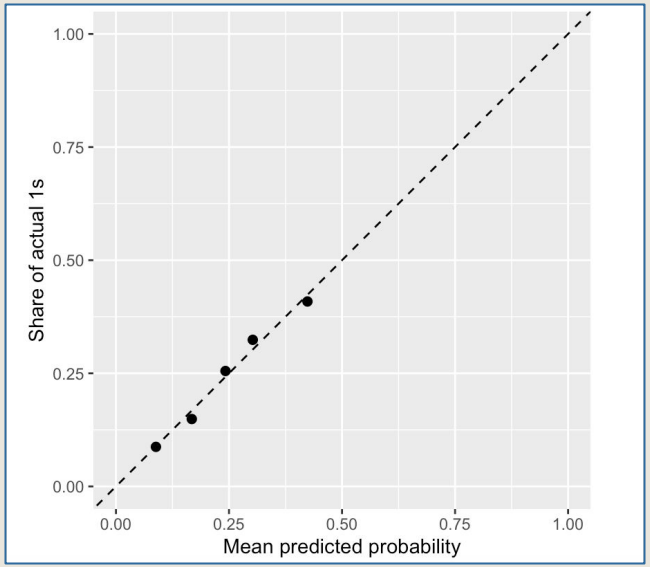
26.8%

predicted chance of signs of mental illness

20 years older raises the predicted chance by about 5 points. Both ages are well inside the observed range of 6–91.

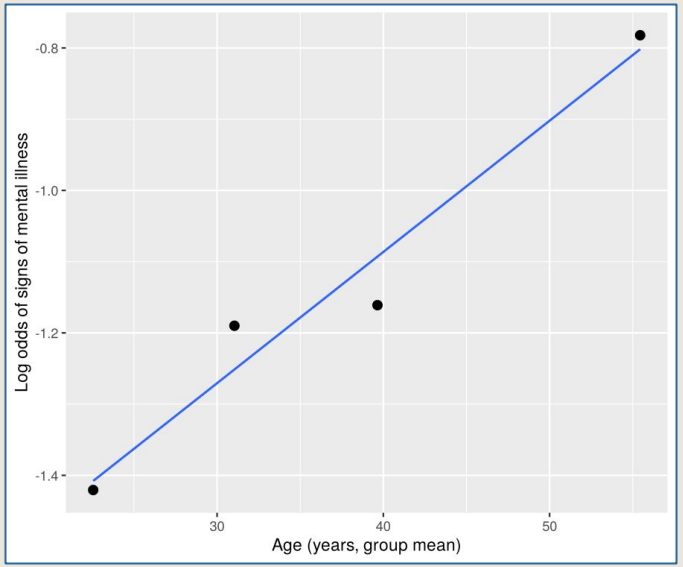
Diagnostic Plots

CALIBRATION



All 5 points sit on the diagonal: predicted rates match observed rates.

LOG ODDS VS AGE



Log odds rise with age along a roughly straight line.

Model Checks

Calibration

In all 5 bins of predicted risk, the observed rate is within about 2 points of the predicted rate, so the points sit near the diagonal.

Log odds vs age

Across 4 age groups the log odds rise from −1.42 to −0.78, flatter between 31 and 40. Roughly straight.

Classification

At a 0.5 cutoff, accuracy is 75.5%, the same as always guessing “no.” Only 28 people are predicted “yes.”

Unusual cases and independence

No errors found: ages 6–91 are all real, and every category has both outcomes, so nothing was removed. Cases may cluster by city.

What We Found

ANSWER

People with **knives or other non-gun objects** had about twice the odds of showing signs of mental illness as people with guns.
People who were **fleeing** had 60–74% lower odds. Odds rise with **age**; **threat level** adds nothing.

LIMITATIONS

- Association, not causation; observational data only
- “Signs of mental illness” comes from news reports and may be under-reported
- Only fatal shootings: says nothing about encounters that ended differently
- Complete-case model drops 1,048 people whose missingness may not be random

References

The Washington Post. *Fatal Force* police shootings database. https://github.com/washingtonpost/data-police-shootings

OpenIntro. *fatal_police_shootings* dataset. https://www.openintro.org/data/index.php?data=fatal_police_shootings